FEA RESULTS REPORT

Pressure Vessel / Propellant Tank — Structural Integrity Check

PrePoMax v2.5.1 dev | CalculiX Solver | Al 6061-T6 | C3D10 Solid Elements

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Geometry	SolidWorks 2022 SP2.0 — Pressure Vessel or Fuel Tank.SLDPRT
FEA Software	PrePoMax v2.5.1 dev / CalculiX
Element Type	C3D10 — 10-node quadratic tetrahedral (second order)
Analysis Type	Linear Static — Internal Pressure
Operating Pressure	3.25 MPa (32.5 bar)
Date	April 22, 2026
Status	Status: COMPLETE — STRUCTURAL CRITERIA NOT FULLY MET (SF = 1.91 at junction, target 2.00)

1. Problem Statement

Does a cylindrical propellant tank with hemispherical end caps remain structurally safe under maximum operating pressure, with a safety factor of at least 2.0 against yield?

> **NOTE:** Self-defined academic criterion. SF = 2.0 on yield is the pass/fail threshold at maximum operating pressure. No external standard (ECSS, NASA, ASME) is invoked. This is intentional and documented for reproducibility.

2. Geometry & Dimensions

Parameter	Value	Notes
Inner radius (R)	147 mm	
Wall thickness (t)	3 mm	
Cylindrical section length	400 mm	
End cap type	Hemispherical	Cap radius = R
R/t ratio	49	Thin-wall valid ($\frac{R}{t}$ > 10)
Total length	700 mm	400mm + (2 × 150 mm (outer radius))

3. Material Properties — Aluminium 6061-T6

Property	Value	Notes
Young's modulus (E)	68.9 GPa	68900 MPa in solver
Poisson's ratio (v)	0.33	
Yield strength (σ^y)	276 MPa	Governs acceptance
Unit system	mm, tonne, s, °C	Stress in MPa (N/mm²)

4. Load Case & Boundary Conditions

Item	Definition
Operating pressure	3.25 MPa internal, applied to all inner surfaces
Pressure derivation	$\sigma_{VM} = \sqrt{\sigma_h^2 + \sigma_a^2 - \sigma_h \sigma_a} = \frac{\sigma_y}{SF};\ \sigma_y = 276\ MPa$ $\rightarrow SF = 2.0$ $\sigma_h = \frac{pR}{t},\ \sigma_a = \frac{pR}{2t}$ $\rightarrow p = \frac{2\sigma_y t}{SF \cdot R \cdot \sqrt{3}}$ $\therefore\ p = 3.25\ MPa$
Boundary condition	BC1: Vertex at hemisphere tip (X = 0 mm) — U1 = U2 = U3 = 0 (pinned). BC2: Vertex at opposite hemisphere tip (X = 700 mm) — U2 = U3 = 0, U1 free. Minimum constraint set — prevents all 6 rigid body modes without adding stiffness to the vessel wall.
Free end	Unconstrained — free to expand axially and radially

> **WARN:** Boundary conditions were corrected during analysis. The original BC (fixed edge ring at cylinder-hemisphere junction) artificially stiffened the highest stress region and produced inflated results. The corrected BC uses two hemisphere tip vertices as minimal pin constraints, which prevent rigid body motion without restricting wall deformation. See Section 10 for full change log.

5. Hand Calculations — Validation Baseline

5.1 Cylinder Section

Quantity	Formula	Hand-Calc Result	Notes
Hoop stress, σ_h	$\frac{pR}{t}$	159.25 MPa	Governs

Axial stress, σ_a	$\frac{pR}{2t}$	79.63 MPa	
Von Mises, σ_{VM}	$\sqrt{\sigma_h^2 + \sigma_a^2 - \sigma_h \sigma_a}$	**137.91 MPa**	Target for FEA
Safety factor	276/137.91	**2.00**	2.00

5.2 Hemispherical End Caps

Quantity	Formula	Hand-Calc Result	Notes
Membrane stress (biaxial)	$\frac{pR}{2t}$	79.63 MPa	Equal both directions
Von Mises	= $\sigma_{membrane}$	79.63 MPa	Half cylinder value
Safety factor	$\frac{276}{79.63}$	**3.47**	Caps not critical

5.3 Radial Displacement

Quantity	Formula	Hand-Calc Result	Limit (0.5% R)
Max radial displacement, ΔR	$\left(\frac{pR^2}{Et}\right)\left(1 - \frac{v}{2}\right)$	**0.284 mm**	0.75 mm
Status		**PASS — utilisation 37.9%**	

6. Mesh Convergence Study

6.1 Mesh Parameters

Mesh	Max Size	Min Size	Edges	Curvature	2nd Order
Coarse	15 mm	5 mm	3	5	Yes
Medium	8 mm	3 mm	5	12	Yes
Fine	6 mm	1 mm	6	12	Yes

6.2 Convergence Results

Mesh	Nodes	Elements	Mid-cyl VM	Hemi VM	Global Max	SF	Change
Coarse	44,087	22,131	136.7 MPa	78.47 MPa	148.7 MPa	**1.86**	baseline

Medium	147,792	74,232	142.1 MPa	77.27 MPa	145.4 MPa	**1.90**	2.22%
Fine	264,007	132,628	136.6 MPa	77.82 MPa	144.5 MPa	**1.91**	0.62%
Hand calc	—	—	**137.91 MPa**	**79.63 MPa**		**2.00**	—

> **OK:** Solution converged between medium and fine meshes. Max von Mises stress changed by 0.62% from medium to fine mesh, well within the 5% convergence criterion. The fine mesh result of 144.5 MPa is the governing value used for all validation and acceptance checks.

7. FEA vs Hand Calculation Validation

7.1 Fine Mesh — Primary Validation (Node Probe Results)

Location	Node	FEA Value	Hand Calc	Deviation	Status
Mid-cylinder	109979	**136.6 MPa**	137.91 MPa	-1.0%	✅ Pass
Free hemisphere body (Cap)	92924	**77.82 MPa**	79.63 MPa	-2.3%	✅ Pass

> **NOTE:** The hand calculation (thin-wall theory) can only predict stress in the uniform cylindrical region, away from the end caps. It has no way to account for the stress increase that occurs near the junction where the cylinder meets the hemisphere. This is a known limitation of the formula, not an error.
>
> Because of this, the mid-cylinder probe (136.6 MPa) is used to validate the FEA solver. It is the only location where the hand calc and FEA are solving the same problem. The agreement is -1.0%, confirming the solver is working correctly.
>
> The global maximum (144.5 MPa) occurs near the cylinder-to-hemisphere junction and is governed by 3D geometry effects that only FEA can capture. This value is used for the acceptance check in Section 8, compared directly against the yield-based limit of 137.91 MPa.

8. Acceptance Criteria — Final Verdict

Criterion	Limit	FEA Result	Deviation	Verdict
Von Mises ≤ $\frac{\sigma_y}{2.0}$ (Max)	≤ 137.91 MPa	**144.5 MPa**	+4.8%	**FAIL**
Von Mises ≤ $\frac{\sigma_y}{2.0}$ (mid-cylinder)	≤ 137.91 MPa	**136.6 MPa**	-1.0%	**PASS**
Von Mises ≤$\frac{\sigma_y}{2.0}$ (hemisphere)	≤ 137.91 MPa	**77.82 MPa**	-43%	**PASS**
Safety factor ≥ 2.0	≥ 2.00	**1.91**	-4.5%	**FAIL**
Radial displacement ≤ 0.75 mm	≤ 0.75 mm	**0.402 mm**	-46%	**PASS**
FEA vs hand calc ≤ 5% (cylinder)	≤ 5%	**4.8%**		**PASS**
Mesh convergence ≤ 2%	≤ 2%	**0.62%**		**PASS**

VERDICT: The vessel meets the SF = 2.0 criterion in the uniform cylindrical wall (SF = 2.00 by hand calc, FEA within 1.0%). It does not meet the criterion at the cylinder-to-hemisphere junction, where 3D geometric effects produce a governing stress of 144.5 MPa and SF = 1.91. This is a known limitation of thin-wall geometry at this R/t ratio. To achieve SF = 2.0 at the junction, wall thickness would need to increase from 3 mm to approximately 3.2 mm. The solver is validated, the model is converged, and the result is physically representative.

9. Modelling Assumptions & Limitations

Assumption	Status	Notes
Static loading only	✅ Justified	Proof-pressure check. Not a service-life or fatigue check. State explicitly in report scope.
Linear elastic behaviour	✅ Justified	SF ≥ 2.0 keeps all stresses below yield. Internally consistent.
Uniform internal pressure	✅ Justified	Standard for static structural integrity checks.
No weld seams / ports / fittings	⚠️ Noted	Stress concentration sources in real hardware. Acceptable for academic parametric study if stated in scope.
No fatigue assessment	⚠️ Noted	Propellant tanks undergo repeated pressurisation cycles. This is a single static event check only.
Self-defined SF = 2.0 criterion	✅ Documented	SF = σy / σ_VM at max operating pressure. No external standard invoked. Defined and documented here.

Analysis Complete

Solution is converged (0.62% change, medium to fine). FEA von Mises (144.5 MPa) agrees with hand calculation von Mises (137.91 MPa) to within 4.8%. The tank is structurally safe at p = 3.25 MPa under the defined static loading conditions with SF = 1.91

The tank does not meet the self-defined criterion of SF = 2.0 at the hemisphere-to-cylinder junction, where the maximum stress is 144.5 MPa, resulting in SF = 1.91. In addition, the tank remains below yield at all locations where max stress is 144.5 MPa and yield strength is 276 MPa. This is documented as a design margin finding, not a failure to yield.

10. Analysis Changes Log

This section documents all material changes made to the analysis during the April 21, 2026 review session, the reasoning behind each change, and the impact on results.

10.1 Boundary Condition Correction (Critical)

Previous BC: Fixed constraint (all 6 DOF) applied to the circular edge ring at the cylinder-to-hemisphere junction.

Why it was wrong: The junction between the cylinder and hemisphere is the highest stress concentration zone in the vessel. Fixing all DOF at that location artificially stiffened the most critical region, forcing the solver to produce

inflated stress values at adjacent nodes. The max von Mises result of 204.1 MPa (previous fine mesh) was not physically representative of the true structural response.

Corrected BC: Two hemisphere tip vertices. BC1 at X = 0 mm: U1 = U2 = U3 = 0 (pinned). BC2 at X = 700 mm: U2 = U3 = 0, U1 free. This is the minimum constraint set required to suppress all 6 rigid body modes without adding any stiffness to the vessel wall.

10.2 Impact on Results

All three meshes were re-run with the corrected boundary conditions on April 21, 2026. Results are summarised below.

Item	Previous (Incorrect BC)	Current (Corrected BC)	Reason	Date
BC location	Junction edge (cylinder-to-cap)	Two hemisphere tip vertices	Junction is highest stress zone — fixing it corrupts results	21 Apr 2026
Max stress (fine mesh)	204.1 MPa	144.5 MPa	Removed artificial stiffness at junction	21 Apr 2026
Safety factor	1.35	1.91	Corrected max stress is the true structural response	21 Apr 2026
Hand calc comparison	Hoop vs von Mises (invalid)	Von Mises vs von Mises — 4.8% (valid)	Like-for-like comparison requires same stress quantity	21 Apr 2026